

Aaditya Sharma

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Available immediately | Part-time, casual, or contract | Unlimited hours until end of July 2026, then 48 hrs/fortnight

Professional Summary

Senior DevOps and MLOps Engineer with over two years of production experience creating CI/CD pipelines, managing cloud infrastructure, and shipping containerised machine learning workloads. Skilled in integrating AI and ML workflows into DevOps lifecycles, from model packaging and versioning through to automated deployment and monitoring of inference services. Experienced with AWS, Kubernetes (EKS), Terraform, and ML pipeline tooling including MLflow and SageMaker Pipelines. Currently pursuing a Master of Information Technology (Artificial Intelligence specialisation) at the University of Melbourne. Immediately available for part-time, casual, or contract engagements and committed to delivering reliable, scalable systems from day one.

Technical Skills

Cloud Platforms: AWS (EC2, S3, Lambda, RDS, IAM, Route 53, SageMaker), Azure (Virtual Machines, Functions)

AI / MLOps: MLflow, AWS SageMaker Pipelines, Kubeflow (basic), model versioning, feature stores, inference monitoring, A/B model deployment, data drift detection, TensorFlow, Keras, Pandas, NumPy

Containers & Orchestration: Docker, Kubernetes (EKS), Helm

CI/CD & Automation: Jenkins, GitHub Actions, Git, GitOps workflows

Infrastructure as Code: Terraform, AWS CloudFormation

Monitoring & Observability: Prometheus, Grafana, ELK Stack, model performance dashboards

Scripting & Databases: Python (advanced), Bash, MySQL, PostgreSQL, MongoDB, Redis

Professional Experience

Senior DevOps Engineer

July 2023 – September 2025

Comviva (Tech Mahindra Company)

- Established CI/CD pipelines using Jenkins and GitHub Actions, cutting deployment time by 40% and enabling automated rollouts for both application and ML model releases.
- Integrated MLflow into deployment workflows for ML model versioning, experiment tracking, and artefact management, supporting data science teams in promoting models from staging to production.
- Containerised ML inference services using Docker and deployed to Kubernetes (EKS) with auto-scaling policies and resource quotas for GPU and CPU usage.
- Configured SageMaker Pipelines for batch inference jobs, lowering manual ML deployment effort by 60% and standardising model promotion workflows across environments.
- Designed Prometheus and Grafana dashboards for real-time model performance monitoring, covering latency, throughput, and data drift alerting for production inference endpoints.
- Provisioned and maintained cloud infrastructure on AWS including EC2, S3, Lambda, RDS, and SageMaker; introduced cost optimisation strategies, cutting monthly spend by approximately \$2,000.
- Provisioned infrastructure using Terraform across development, staging, and production environments, maintaining reusable modules and enforcing infrastructure-as-code best practices.
- Set up ELK Stack centralised logging for diagnosing model serving errors and pipeline failures in production, shortening mean time to resolution.
- Streamlined operational workflows using Python scripts and AWS Lambda, saving over 15 hours per week across the team.
- Collaborated closely with development and data science teams to optimise system performance, lower model inference latency, and improve pipeline reliability.
- Mentored and onboarded interns and new team members, shortening ramp-up time by 30%, and conducted knowledge transfer sessions to standardise DevOps and MLOps practices across the organisation.

Projects

- **RAG-Ops: Production-Grade Document Intelligence Platform.** Engineered an end-to-end Retrieval-Augmented Generation system enabling users to upload any PDF and query it conversationally, backed by a full MLOps lifecycle. Designed a RAG pipeline using LangChain, ChromaDB for vector storage, and a locally-hosted Llama 3.2 model via Ollama, served through a FastAPI backend. Containerised with Docker and rolled out to a Kubernetes cluster with two replicas, readiness and liveness probes. Instrumented every ingestion and query as an MLflow experiment run, capturing latency, chunk counts, and retrieval metrics. Established real-time observability with Prometheus metrics and a Grafana dashboard covering query throughput, p95 latency, and error rates. Configured testing, linting, and Docker image builds on every push using a GitHub Actions CI/CD pipeline.
- **MindVault: Multimodal Mental Wellness Monitoring System.** Designed and built a passive longitudinal wellness platform that analyses both acoustic and semantic signals from daily voice check-ins. Used faster-whisper for local speech transcription, librosa to extract pitch, RMS energy, MFCCs, and zero-crossing rate, and DistilBERT (HuggingFace) for sentiment classification. Fused both signals into a composite mood score with divergence detection, flagging sessions where voice energy and word sentiment disagree, a clinically meaningful signal. Built a FastAPI backend with SQLite persistence, a Chart.js trend dashboard, and a research export endpoint. Logged every check-in as an MLflow experiment run; exposed Prometheus metrics including mood score distributions, pitch gauge, and processing latency visualised in Grafana. Containerised with Docker and deployed on Kubernetes. Designed with a research-ready architecture suitable for longitudinal mental health studies.
- **Medical Image Segmentation.** Developed CNN-based segmentation models using TensorFlow and Keras; packaged models into Docker containers and explored deployment via REST API endpoints.

Education

Master of Information Technology (AI Specialisation) 2026 – Present
University of Melbourne

Coursework focus: Machine Learning, Deep Learning, Cloud Computing, AI Ethics
COMP90038 Algorithms and Complexity — First Class Honours (H1), Semester 1 2026

Bachelor of Technology – Information Technology 2019 – 2023
Manipal University Jaipur

GPA: 8.74/10 | Dean’s List – Highest GPA in 7th Semester (April 2023)

Awards & Recognition

- **Dean’s List – Academic Excellence.** Awarded for achieving the highest GPA in the 7th Semester, Information Technology, Manipal University Jaipur (April 2023).
- **ACE Certificates (Appreciation for Commitment and Effort).** Multiple awards from senior managers and directors at Comviva in recognition of outstanding delivery, initiative, and team contribution.